Python Lists Cheat Sheet

A beginner-friendly reference guide to sequences & arrays in Python

PYTHON 3.X

1. WHAT IS A LIST?

A **List** is an ordered, changeable (mutable) collection of items enclosed in square brackets `[]`. Lists can hold mixed data types.

```
# Creating lists empty_list = []
fruits = ["apple", "banana", "cherry"]
numbers = [1, 2, 3, 4, 5] mixed = ["Alex", 15, True, 9.5]
```

2. ACCESSING ELEMENTS (INDEXING)

Indices start at **0**. Use negative indices to count backwards from the end (starting at **-1**).

Pos Index	0	1	2	3
Value	"A"	"B"	"C"	"D"
Neg Index	-4	-3	-2	-1

3. LIST SLICING: `LIST[START:STOP:STEP]`

Extract a sub-range of items. The `stop` index is *excluded*!

```
nums = [10, 20, 30, 40, 50]
nums[1:4] # [20, 30, 40] (index 1 to 3)
nums[:3] # [10, 20, 30] (first 3 items)
nums[2:] # [30, 40, 50] (index 2 to end)
nums[::-1] # [50, 40, 30, 20, 10] (Reversed)
```

4. ADDING & REMOVING ITEMS

Method	Description & Example
`append(x)`	Adds `x` to the **end** of the list. `nums.append(60)`
`insert(i, x)`	Inserts `x` at index `i`. `nums.insert(0, 5)`
`pop(i)`	Removes & returns item at index `i` (default: last item).
`remove(x)`	Removes first occurrence of value `x`.
`clear()`	Removes all items from list.

5. ESSENTIAL LIST FUNCTIONS & METHODS

Function	What it does
`len(list)`	Returns the number of elements in list.
`min(list)` / `max(list)`	Returns smallest or largest item.
`sum(list)`	Returns total sum of numeric items.
`x in list`	Returns `True` if `x` exists in list.

Method	What it does
`sort()`	Sorts list in place (ascending order).
`reverse()`	Reverses the order of items in place.
`count(x)`	Counts how many times `x` appears.
`index(x)`	Returns the index of the first `x`.

6. LOOPING OVER LISTS

```
# Direct iteration for fruit in fruits: print(fruit) # Loop with index numbers for i, fruit in enumerate(fruits): print(f"{i}: {fruit}")
```

7. LIST COMPREHENSION (SHORT SYNTAX)

Create new lists from existing ones in 1 line!

```
# Syntax: [expression for item in iterable if condition] numbers = [1, 2, 3, 4, 5] squares = [n**2 for n in numbers] # Result: [1, 4, 9, 16, 25] evens = [n for n in numbers if n % 2 == 0] # Result: [2, 4]
```

💡 **Student Pro-Tips:**

- **IndexError:** Occurs when you try to access an index that doesn't exist (e.g., `my_list[len(my_list)]`). Remember, max index is `len - 1`!
- **Modifying vs Copying:** `sort()` changes the original list, while `sorted(list)` returns a new sorted copy.